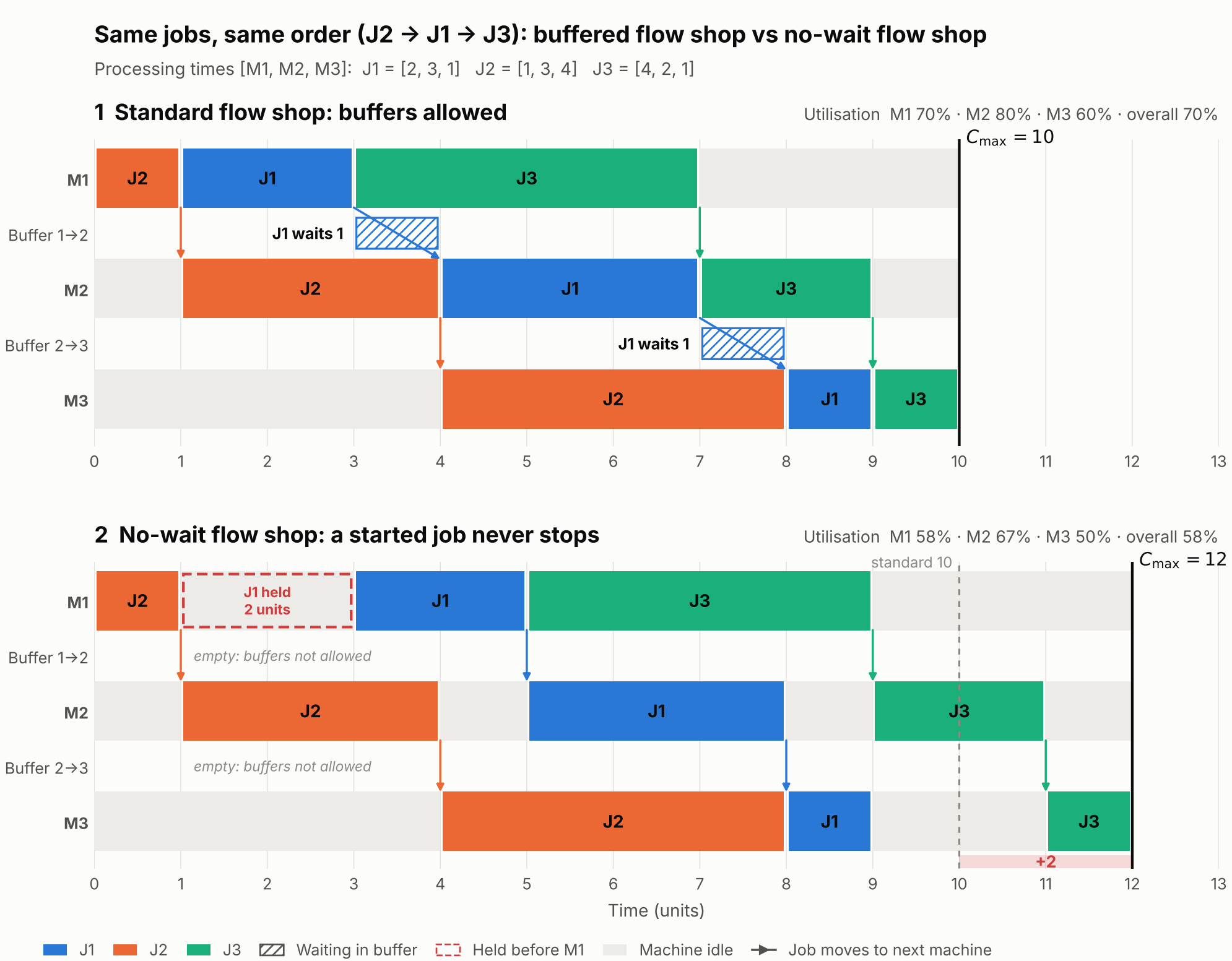
Same jobs, same order (J2 → J1 → J3): buffered flow shop vs no-wait flow shop
Processing times [M1, M2, M3]: J1 = [2, 3, 1] J2 = [1, 3, 4] J3 = [4, 2, 1]
1 Standard flow shop: buffers allowed
Utilisation M1 70% · M2 80% · M3 60% · overall 70%
Cmax = 10
M1
Buffer 1→2
M2
Buffer 2→3
M3
J2
J1
J3
J1 waits 1
J1 waits 1
0 1 2 3 4 5 6 7 8 9 10 11 12 13
2 No-wait flow shop: a started job never stops
Utilisation M1 58% · M2 67% · M3 50% · overall 58%
standard 10
Cmax = 12
J1 held
2 units
empty: buffers not allowed
empty: buffers not allowed
+2
0 1 2 3 4 5 6 7 8 9 10 11 12 13
Time (units)
J1 J2 J3 Waiting in buffer Held before M1 Machine idle Job moves to next machine